

Exploring Pascal's Triangle

GroupSolve

September 19th, 2026

Homework

Introduction

This week, we investigated the Pascal's Triangle. In doing so, we learned about the famous Lucas's Theorem and Kummer's Theorem. These allow us to investigate the binomials modulo a certain number further. In this problem set, we will get a better grasp on the binomials through their properties as well as learn about the Lucas's Theorem and Kummer's Theorem more. As a bonus, we will explore some of the questions from the final slide of the lecture at the end.

Feel free to do any of the problems. I really do not care how much you do. As long as you give your best attempt on the problems you attempt. After all, I want you to discover properties! I am just here to guide you, support you, and provide feedback. There really is no minimum amount of problems you have to do. With that being said, there also is no maximum! Feel free to explore continuations of these problems – anything you find – and submit them to me for feedback! In addition, I request only submitting problems you do. Feel free to search online and consult resources, but I humbly request that you only submit solutions to problems you do by yourself.

If you do have questions, which I do expect you will have many as you solve, feel free to contact me. You can also ask me for any tips, advice, and hints! In general, if you want any help with literally anything, including unrelated things to this problem set, please reach out! I promise I am kind and I want to help!

Submit your problems to this link. The due date is 8/30. Good luck!

– Jacob Khohayting

Binomials

For a lot of the combinatorial identity problems, there are two ways you can prove things (generally). Either algebraically or combinatorially.

1. Compute $0!, 1!, 2!, \dots, 9!, 10!$.
2. Compute $\binom{7}{3}, \binom{8}{4}, \binom{10}{5}$.
3. Find the x^3 coefficient of $(2x + 3)^5$. Generalize and find the x^k coefficient of $(ax + b)^n$.
4. (Pascal's Identity) Show that $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.
5. (Chairman's Identity) Show that $k\binom{n}{k} = n\binom{n-1}{k-1}$.
6. Show that

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

7. Show that

$$\sum_{k=0}^n k\binom{n}{k} = n2^{n-1}.$$

8. Determine a closed form for

$$\sum_{k=0}^n k^2 \binom{n}{k}.$$

9. (Hockey Stick Identity) A committee of $k+1$ people is chosen from a group of $n+1$ people. By conditioning on the person with the largest identification number in the committee, give a combinatorial proof of the Hockey Stick Identity to show

$$\sum_{i=k}^n \binom{i}{k} = \binom{n+1}{k+1}$$

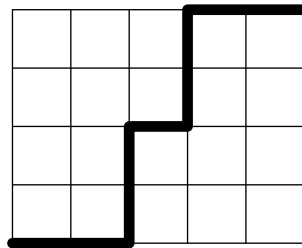
10. Calculate the value of $\sum_{i=3}^{10} \binom{i}{3}$.

11. Show that

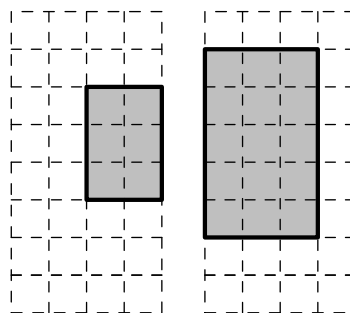
$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

(Hint: Count subsets of $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\}$.)

12. Count the number of lattice paths from $(0,0)$ to $(5,4)$. One example is shown below:



13. Count the number of rectangles that can be formed using the edges of a 4×8 grid. Here are some examples:



Binomials (mod m)

- Express $n = 2026$ and $k = 100$ in base 3.
- Use Lucas's Theorem to compute $\binom{2026}{100} \pmod{3}$.
- Let p be a prime. How many entries in row n of Pascal's triangle are not divisible by p ? Calculate the number of entries in row $n = 100$ that are not divisible by 5.

According to Kummer's Theorem, $\nu_p \left(\binom{n}{k} \right)$ equals the number of carries when adding k and $n-k$ in base p .

4. Add 1101_2 and 1001_2 in base 2. Add 436_7 and 5432_7 in base 7. (Think about how you add in base 10, it is the same!)
5. Find the exact power of 2 dividing $\binom{31}{10}$.
6. Determine the smallest positive integer n such that row n contains an entry divisible by $2^3 = 8$.

Here is a tougher set of problems! They are some of my favorite applications of the theorems you may have just learned!

7. Compute the number of non-negative integers $k < 2^{20}$ such that $\binom{5k}{k}$ is odd.
8. Find the sum of all prime numbers p such that $\binom{20262026p}{p} \equiv 2026 \pmod{p}$.
9. Let m be a positive integer and $f(m)$ be the greatest k such that 2^k divides m . Prove that there exist infinitely many numbers m such that $m - f(m) = 2026$.
10. Let p be a prime and n be a positive integer. Then, p does not divide

$$\prod_{k=0}^n \binom{n}{k}$$

if and only if n can be expressed as $p^s \cdot m - 1$ for $1 \leq m < p$.

11. Prove that $\binom{2n}{n}$ divides $\text{lcm}(1, 2, 3, \dots, 2n)$.
12. Show that the sequence

$$\binom{2026}{2026}, \binom{2027}{2026}, \binom{2028}{2026}, \dots$$

taken modulo 2026, is periodic.

Sierpinski's Triangle

This section is more for exploration purposes! Have fun!

1. Let p be prime. Consider what happens if you take the Pascal's Triangle mod p where p is 3, 5, or 7.
 - (1) Give an element one color if it is $0 \pmod{p}$ and another color if an element is not $0 \pmod{p}$.
 - (2) Give a color if an element is $0 \pmod{p}$, $1 \pmod{p}$, $2 \pmod{p}$, and so on.
 (Tip: Look at examples, formulate patterns, and try to prove the formulations.)

2. Show that

$$\binom{pa}{pb} \equiv \binom{a}{b} \pmod{p}.$$

3. Show that

$$\binom{pa}{pb} \equiv \binom{a}{b} \pmod{p^2}.$$

4. Investigate

$$\binom{2a}{2b}, \binom{2a+1}{2b}, \binom{2a}{2b+1}, \binom{2a+1}{2b+1} \pmod{4},$$

in relation to $\binom{a}{b} \pmod{4}$.

5. Consider what happens if you take the Pascal's Triangle modulo 4. Give an element one color if it is odd, another color if it is even but not divisible by 4, and another color if it is divisible by 4. Can you try this for more prime powers like 9?